Class-2

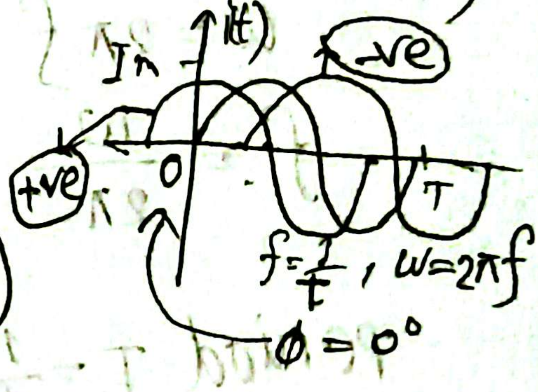
## · General format for sinusoidal current and voltage

$i(t) = I_m \sin(\omega t + \varphi) = I_m \sin\alpha$, $i(t) = I_m \cos(\omega t + \varphi)$

$v(t) = V_m \sin(\omega t + \varphi) = V_m \sin\alpha$, $v(t) = V_m \cos(\omega t + \varphi)$

Q) $v(t) = 100 \sin(2\pi 100t - \frac{\pi}{6})$

$v(t) = 100 \sin(2\pi 100t - 30^\circ)$




1] Find the amplitude, phase, period, and frequency of the sinusoid.

$$v(t) = 12\cos(50t + 10^\circ)$$

2] a. Determine the angle at which the magnitude of the sinusoidal function $v = 10\sin(377t)$ is 4V.

b] Determine the time at which the magnitude is attained.

Ans → $v(t) = \cancel{10}\ 12\cos(50t + 10^\circ)$ volt

$V_m \cos(\omega t + \phi)$ volt

amplitude $= 12V$

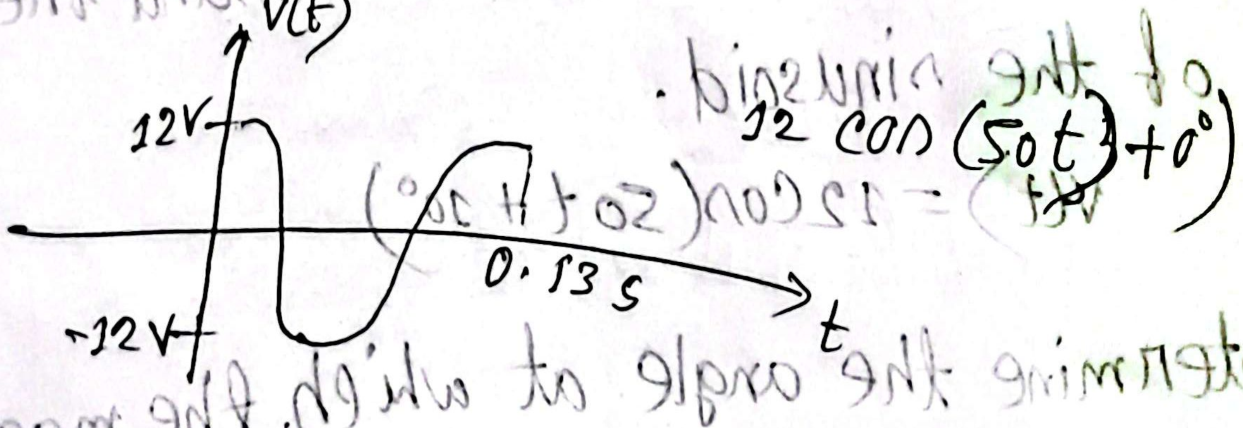
phase $= 10^\circ$

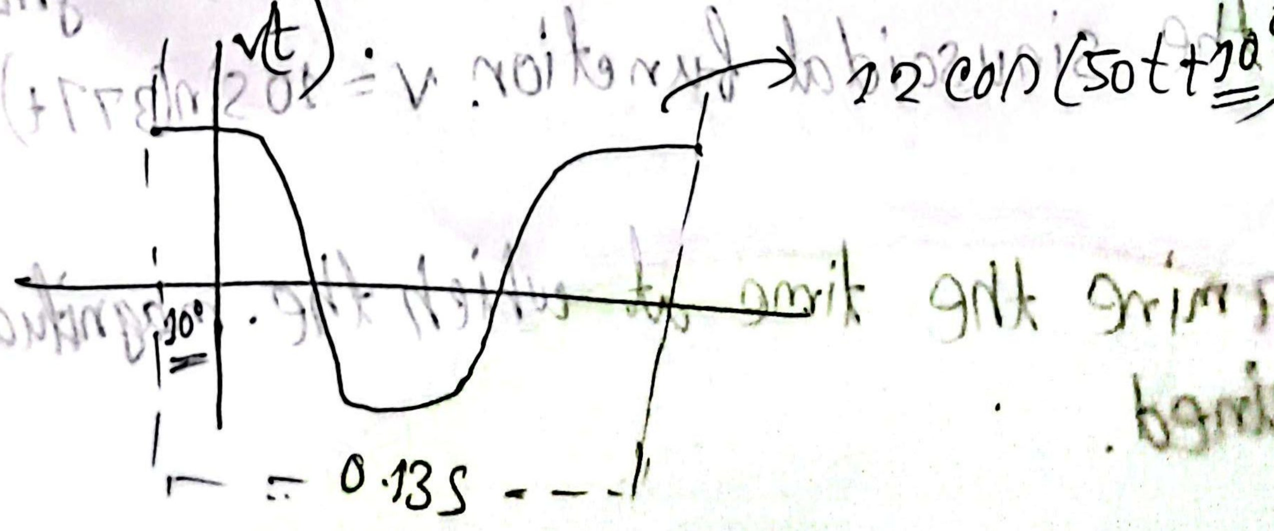
angular frequency $\omega = 50$ rads$^{-1}$

$\omega = 2\pi f$

$f = \frac{\omega}{2\pi} = 7.96$ Hz

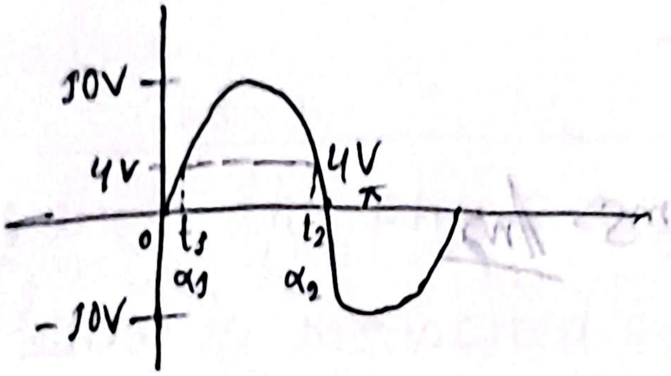
period $T = \frac{1}{f} = 0.13$ s



Ans

2] a] $v = 10\sin(377t)$ volt




now,

$$v(t) = 10\sin(377t)$$

$$\Rightarrow 4 = 10\sin(377t_1)$$

$$\Rightarrow \frac{4}{10} = \sin(377t_1)$$

$\boxed{\text{In radian mode in calculator}}$

$$\Rightarrow t_1 = \frac{1}{377}\sin^{-1}\left(\frac{4}{10}\right)$$

$$= 1.09\,\text{ms}$$

again,

$$w = \frac{\alpha_1}{t_1}$$

$$\therefore \alpha_1 = w \times t_1$$

$$= 377 \times 1.09 \times 10^{-3}$$

$$= 0.412\text{ rad.}$$

angle $\alpha_2 = \pi - \alpha_1$

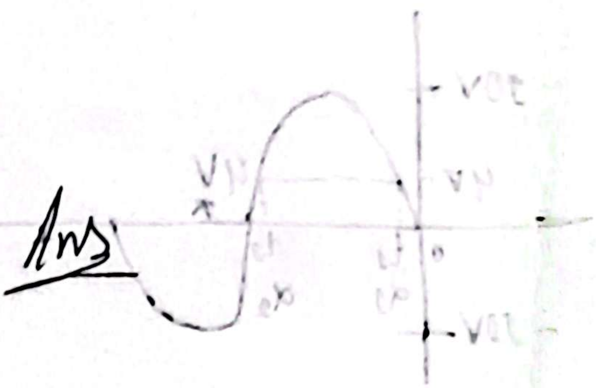
$= 2.72$ rad.

Ans

b) $\omega = \dfrac{\alpha_2}{t_2}$

$\Rightarrow t_2 = \dfrac{\alpha_2}{\omega} = \dfrac{2.72}{377}$

$= 7.24$ ms. Ans



## · Lagging and leading

Two sinusoidal waves whose phases are to be compared must :-

1) Both be written as sine waves, or both as cosine waves.

2) Both be written with positive amplitudes.

3) Each have the same frequency.

$-\sin\alpha = \sin(\alpha \pm 180^\circ)$

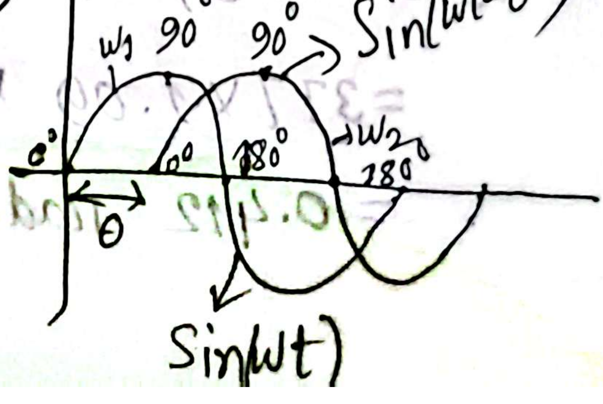
$-\cos\alpha = \cos(\alpha \pm 180^\circ)$

$\mp \sin\alpha = \cos(\alpha \pm 90^\circ)$

$\pm \cos\alpha = \sin(\alpha \pm 90^\circ)$

$w_1$ leading $w_2$

$w_2$ lagging $w_1$



Q] calculate the phase angle between $v_1 = -10\cos(\omega t + 50^\circ)$ and $v_2 = 12\sin(\omega t - 10^\circ)$. State which sinusoidal is leading.

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<u>Notes</u> that the phase angle between the two waveforms is measured between those two points on the horizontal axis through which each passes with the same slope.

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<u>Solve</u>

$$v_1 = -10\cos(\omega t + 50^\circ) \text{ volt}$$

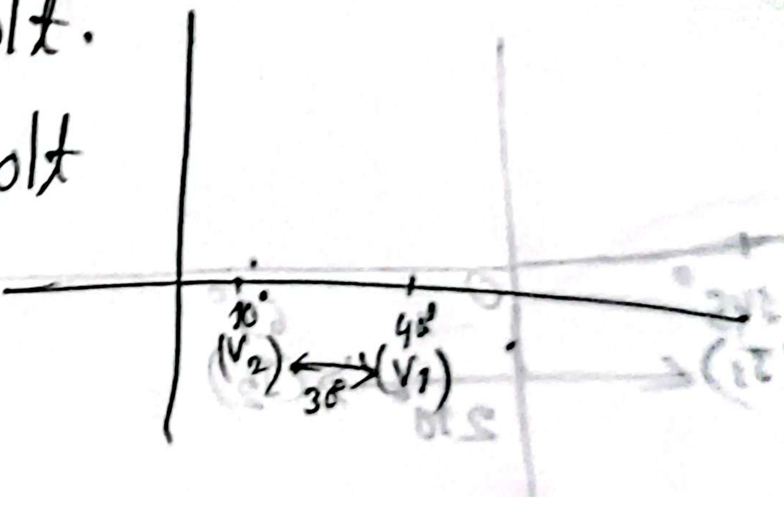
$$v_2 = 12\sin(\omega t - 10^\circ) \text{ volt}$$

$$v_1 = -10\cos(\omega t + 50^\circ)$$

$$= 10\sin(\omega t + 50^\circ - 90^\circ) \quad [-\cos a = \sin(a - 90^\circ)]$$

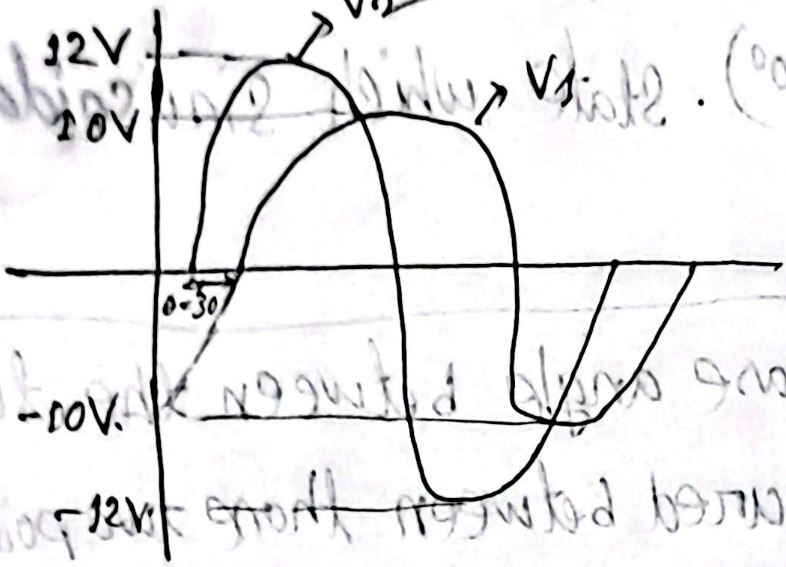
$$= 10\sin(\omega t - 40^\circ) \text{ volt.}$$

$$v_2 = 12\sin(\omega t - 10^\circ) \text{ volt}$$




$v_2$ leads $v_1$ by $30^\circ$

$v_1$ lags $v_2$ by $30^\circ$




Q1 $i_1 = -4\sin(377t + 55^\circ)$ Phase angle difference

$i_2 = 5\cos(377t - 65^\circ)$

Ans

$i_1 = -4\sin(377t + 55^\circ)$

$= 4\cos(377t + 55^\circ + 90^\circ)$ [3 number transform]

$= 4\cos(377t + 145^\circ)$

$i_2 = 5\cos(377t - 65^\circ)$

$i_1$ leads $i_2$ by $210^\circ$

$i_2$ lags $i_1$ by $210^\circ$